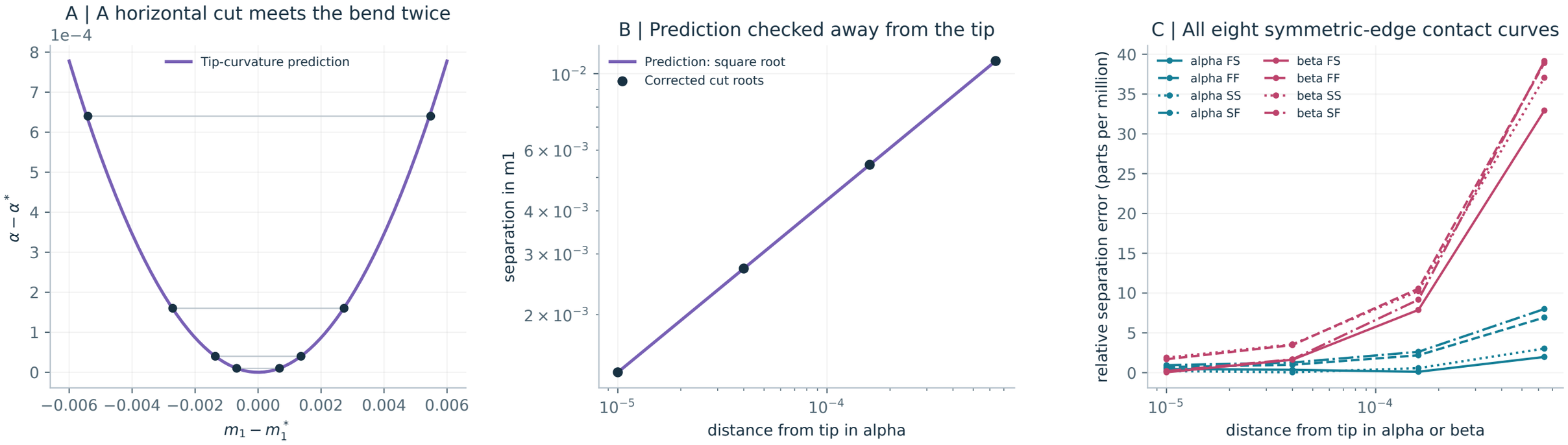
The geometry predicts where the two islands will appear

$\Delta m_1 \simeq 2\sqrt{2\delta/|a''(m_1^*)|}$ | curvature is measured at the tip, not fitted to the cut roots



32 held-out horizontal cuts; maximum measured separation error below 0.004%. F = fold; S = symmetry loss.
This is a standard local geometric square-root law, verified here for these contact curves; the estimates are not interval-certified.